

SANTA MARIA, CA, USA

WMO: 723940

Lat: 34.899N

Lon: 120.449W

Elev: 242

StdP: 14.57

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23273

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	33.6	35.9	21.2	16.0	55.6	25.0	19.3	50.7	23.7	58.6	20.5	58.1	2.9	120	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	20.3	85.1	62.3	80.9	61.8	77.4	61.3	66.5	78.4	65.0	76.2	63.8	74.2	9.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	84.4	69.8	60.5	79.4	68.5	59.3	76.1	67.3	31.2	78.5	30.0	76.0	29.1	74.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.4	20.6	18.3	28.9	95.0	2.5	4.6	27.1	98.2	25.7	100.9	24.3	103.5	22.5	106.8	
			26.8	69.4	2.5	2.4	25.0	71.1	23.6	72.5	22.2	73.9	20.4	75.7	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	52.9	53.4	55.4	56.3	58.7	61.0	63.5	63.9	64.2	61.9	57.0	52.4
	DBStd	6.20	4.99	5.31	4.86	4.90	3.90	3.40	3.15	3.08	4.26	5.23	5.43	4.85
	HDD50	102	27	22	9	6	0	0	0	0	0	0	7	31
	HDD65	2554	374	327	301	265	203	127	67	58	65	127	248	392
	CDD50	3170	118	116	175	196	268	330	420	432	425	370	216	104
	CDD65	148	0	1	2	5	7	8	22	24	40	32	7	0
	CDH74	1700	30	50	67	112	122	109	135	146	351	427	135	16
	CDH80	482	2	9	13	36	41	33	18	30	114	154	32	0
(14) Wind	WSAvg	6.6	5.8	6.6	7.3	8.1	8.4	7.6	6.4	6.0	5.8	5.9	5.6	5.7
(15) Precipitation	PrecAvg	12.6	2.4	2.8	2.3	0.9	0.2	0.0	0.0	0.0	0.2	0.6	1.3	1.9
	PrecMax	26.8	11.8	11.6	9.4	3.9	2.4	0.9	0.4	0.9	1.6	2.4	4.7	9.8
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.4	2.3	2.7	2.0	1.1	0.4	0.1	0.1	0.1	0.3	0.6	1.2	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.2	80.7	81.0	84.8	85.3	83.3	82.3	83.5	90.4	91.6	84.4	76.2
		MCWB	55.1	59.1	60.1	60.6	61.2	65.3	66.4	65.7	64.8	63.2	60.4	54.3
	2%	DB	72.2	73.5	74.4	76.1	76.1	76.2	77.5	77.4	82.6	84.2	77.7	70.8
		MCWB	55.0	56.9	57.7	58.7	60.6	62.5	64.7	64.8	63.9	60.1	57.4	54.3
	5%	DB	67.9	67.9	69.8	70.7	70.9	72.3	74.3	74.4	77.3	78.3	72.6	66.2
		MCWB	54.0	55.3	56.6	57.9	59.0	61.2	63.5	63.5	63.5	59.5	56.4	53.6
	10%	DB	63.2	63.4	65.5	66.6	67.5	69.7	71.6	72.1	73.4	73.1	68.0	62.6
		MCWB	53.8	53.9	55.8	56.3	57.8	59.8	62.0	62.4	62.4	59.2	55.9	53.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.3	62.1	63.0	63.8	65.0	65.5	68.2	67.8	69.8	67.4	63.0	61.2
		MCDB	67.4	72.5	75.0	78.3	78.4	79.7	79.0	80.0	82.1	81.5	75.0	67.1
	2%	WB	58.8	59.3	60.3	60.7	61.9	63.4	65.5	65.4	66.4	64.3	60.8	58.8
		MCDB	64.9	67.8	70.0	72.1	73.6	74.7	76.5	76.1	77.6	75.9	70.3	63.5
	5%	WB	57.0	57.4	58.5	58.9	59.7	61.8	63.9	64.0	64.3	62.3	59.2	56.9
		MCDB	62.6	65.0	66.9	69.1	69.4	71.8	73.6	73.9	74.9	73.1	67.7	61.5
	10%	WB	55.2	55.4	56.8	57.0	58.2	60.1	62.3	62.5	62.7	60.4	57.6	55.0
		MCDB	61.2	61.8	64.1	65.3	66.6	68.7	71.0	71.4	72.2	70.2	65.9	60.3
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	21.0	21.2	21.6	19.5	18.7	17.7	17.9	20.3	23.1	23.6	22.4
		MCDBR	29.6	29.2	28.8	29.2	26.2	23.3	21.3	21.0	27.0	32.7	30.9	28.4
	5% WB	MCWBR	16.3	16.1	15.5	15.2	13.3	11.5	10.3	10.0	12.6	15.2	15.3	16.0
		MCWBR	20.9	22.9	23.1	24.9	22.6	21.7	20.1	20.0	23.2	25.6	24.0	19.1
(40) Clear-Sky Solar Irradiance	taub		0.304	0.320	0.324	0.341	0.352	0.352	0.361	0.347	0.347	0.342	0.304	0.308
		taud	2.583	2.543	2.537	2.476	2.465	2.461	2.490	2.497	2.503	2.517	2.599	2.583
	Ebn at Noon		289	295	302	299	295	294	290	294	290	283	286	280
		Edh at Noon	24	28	31	35	35	36	34	33	32	29	24	23
(44) All-Sky Solar Radiation	RadAvg		945	1219	1669	2087	2343	2447	2326	2151	1846	1411	1065	844
	RadStd		75	126	112	131	140	102	77	79	82	96	58	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.13	N/A	N/A	+1.85	N/A	N/A	N/A	-385	+425
(47) Regional (27 neighbors)		+1.17	N/A	-2.30	+1.35	N/A	N/A	N/A	-301	+432

Nomenclature: See separate page